

Automatic DC-Offset Calibration for a PSoC-Based Geophone Front End

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Abstract—Accumulated amplifier and reference offsets can saturate intermediate nodes in geophone front ends built around a programmable system-on-chip (PSoC). This paper presents a foreground method and proof-of-concept evaluation that sequentially recenters the programmable-gain amplifier (PGA), band-pass filter (BP), summing stage (SUM), and low-pass filter (LP) using on-chip conversion and filtering resources. On two independently assembled boards and reported PGA settings spanning $\times 1$ – $\times 50$, oscilloscope measurements show LP sample-mean offsets at the node feeding the analog-to-digital converter (ADC) of 0.4–3.9% of the nominal 5-V full-scale span across eight board/gain combinations, versus 2.9–45.8% under a shared fixed-reference baseline. The larger PGA residual is consistent with its wide, non-optimized acceptance band. An unlogged stationary-bench observation placed run completion near 10 s. A multichannel surface-wave record demonstrates field operation, but is not a controlled baseline comparison.

Index Terms—Analog front end, DC-offset calibration, geophone, mixed-signal systems, programmable system-on-chip (PSoC)

I. INTRODUCTION

Multichannel surface-wave methods use distributed vibration measurements to estimate Rayleigh-wave dispersion and infer near-surface mechanical properties [1], [2]. Although their processing is well established, acquisition hardware remains a major source of cost and deployment complexity. Low-cost geophone acquisition has been demonstrated with embedded mixed-signal hardware [3], and embedded geophone interfaces combine instrumentation amplification, filtering, and resistor-adjustable gain [4]. Achieving low offset through higher-grade analog components, however, raises the cost of every node. This work instead uses resources already present in the microcontroller to correct the operating point of a budget-constrained front end.

The system belongs to a low-cost, reconfigurable Internet-of-Things acquisition platform [5] based on SM-24 moving-coil geophones [6]. Its programmable system-on-chip (PSoC) 5LP integrates programmable-gain amplifiers (PGAs), operational amplifiers, analog filters, voltage digital-to-analog converters (VDACs), multiplexers, a delta-sigma analog-to-digital converter (ADC), a digital filter, direct memory access (DMA), and an Arm Cortex-M3 processor [7]. Time-sharing the ADC and digital filter between acquisition and calibration avoids a separate converter or processor, following digitally assisted analog design [8]. The tradeoff is that the internal analog blocks cannot be independently selected to optimize noise and offset,

a limitation also reported for a single-chip PSoC potentiostat [9].

The conditioning circuit derives from the cascaded geophone front end of Ma *et al.* [10]. On the PSoC, accumulated DC errors consumed operating range at intermediate and final nodes; a finite foreground calibration recenters them because a continuous servo proved impractical (Section V).

Prior PSoC calibration estimated and stored numerical transimpedance-amplifier, PGA, and ADC corrections under internal stimulus [9], [11], while body-bias calibration physically trimmed one amplifier [12]. Here, the loop instead adjusts the physical reference of each trim-accessible node within the available VDAC range and resolution.

The sensor-to-PGA and direct-SUM paths remain DC-coupled because placing a blocking corner ahead of the instrumentation input well below 1 Hz would require a large-value off-chip coupling capacitor. Per-stage trimming addresses offset without introducing that component or classical autozero/chopper circuitry [13].

The objective is to establish the feasibility of recentring accumulated DC offsets with PSoC-resident sensing, filtering, and VDAC actuation, without added calibration circuitry, and to quantify preliminary performance against a shared fixed-reference baseline rather than per-condition optimized codes. The contributions are: 1) a four-VDAC per-stage architecture reusing the PSoC multiplexer, ADC, and digital filter; 2) a bounded deadband controller that freezes its codes before acquisition; and 3) a two-board proof-of-concept evaluation over PGA settings $\times 1$ – $\times 50$ with one controller-parameter set. Sections II–III present the architecture and controller, Section IV the experiments, and Sections V and VI the interpretation and conclusions.

II. PLATFORM DESIGN

The PSoC conditions and digitizes the geophone output in acquisition mode. In calibration mode, its measurement path, digital-filter coefficients, and DAC-control firmware form a temporary mixed-signal loop without changing the sensor input chain.

In the feed-forward signal chain of Fig. 1, an offset introduced at any stage can propagate through downstream gains toward the ADC input. The path is not a strict one-input cascade: the band-pass filter (BP) follows the programmable-gain amplifier (PGA), whereas the summing stage (SUM) combines a direct PGA branch with the BP branch before

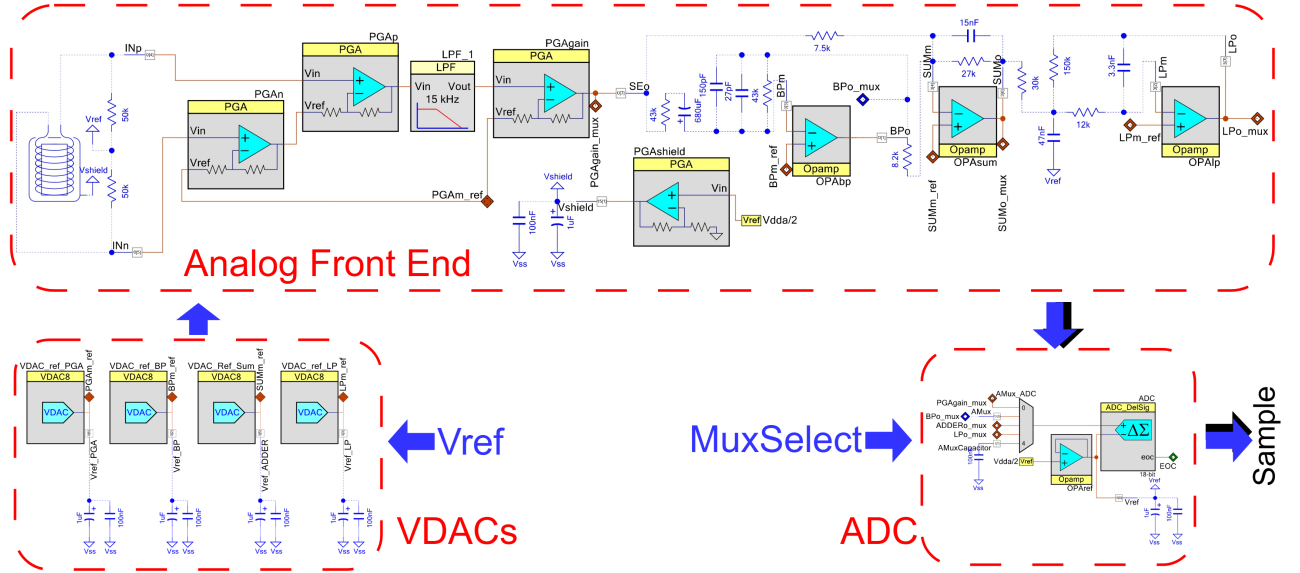


Fig. 1. Analog signal chain: four independent VDAC references trim the programmable-gain amplifier (PGA), band-pass filter (BP), summing stage (SUM), and low-pass filter (LP), with a shared ADC/multiplexer measurement path.

the low-pass filter (LP). The calibration therefore exposes and trims each stage separately rather than correcting only the final ADC value.

An analog multiplexer connects one stage output at a time to the shared 18-bit delta-sigma ADC, which serves as the error sensor; software mutual exclusion prevents simultaneous connections. A switched capacitor behind the selected multiplexer MOSFET combines with its on-resistance for limited RC anti-aliasing and isolates the amplifier from a direct capacitive load.

For the i th measured node, let y_i^{DC} denote its slowly varying voltage deviation from the intended stage reference, with polarity matching the differential ADC coordinates. It can be represented as

$$y_i^{\text{DC}} = \sum_{j < i} A_{ij} y_j^{\text{DC}} + b_i + K_i u_i, \quad (1)$$

where A_{ij} is the DC coupling from an upstream node j (zero when no such path exists), b_i collects amplifier and reference errors in voltage coordinates, u_i is the VDAC voltage, and K_i is the signed DC gain from that VDAC to the measured node. The firmware represents u_i by an 8-bit command code c_i , such that $u_i = q_{\text{DAC}} c_i$. The multiple-input form covers both ordinary cascaded stages and nodes such as SUM; the calibration target is $y_i^{\text{DC}} = 0$.

The architecture assigns one u_i to each trim-accessible node. With negligible reverse coupling, sensor and filter-state changes small relative to the selected deadband during DC estimation, and adequate VDAC authority and range, Eq. 1 motivates a forward pass that brings each measured y_i^{DC} toward zero in PGA, BP, SUM, and LP order (Section III). Equation 1 is an equilibrium model; it does not by itself guarantee finite-time settling of capacitor transients between stages.

III. SELF-CALIBRATING GEOPHONE PLATFORM

A. Digital Hardware and Calibration Sequence

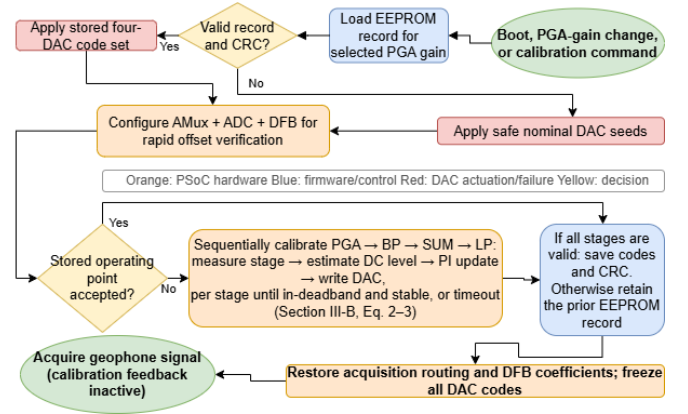


Fig. 2. Foreground calibration flow: stored-code verification, sequential trimming, and acquisition.

At boot, after a gain change, or on command, the firmware loads stored DAC codes and verifies the offsets (Fig. 2). Since DC estimation assumes a stationary geophone and sufficiently slow internal-state change over the lock window, an accepted operating point returns directly to acquisition; otherwise PGA, BP, SUM, and LP are processed sequentially, with finite-impulse-response (FIR) filter reset and settling after each switch. A watchdog and sample limits bound the sequence. During acquisition the controller is inactive, LP is selected, and the VDAC codes remain constant.

The shared digital filter implements the geophone response during acquisition and a 128-tap low-pass FIR DC estimator during calibration (Fig. 3); its state and acquisition path are restored afterward. A gain-indexed electrically erasable programmable read-only memory (EEPROM) record stores

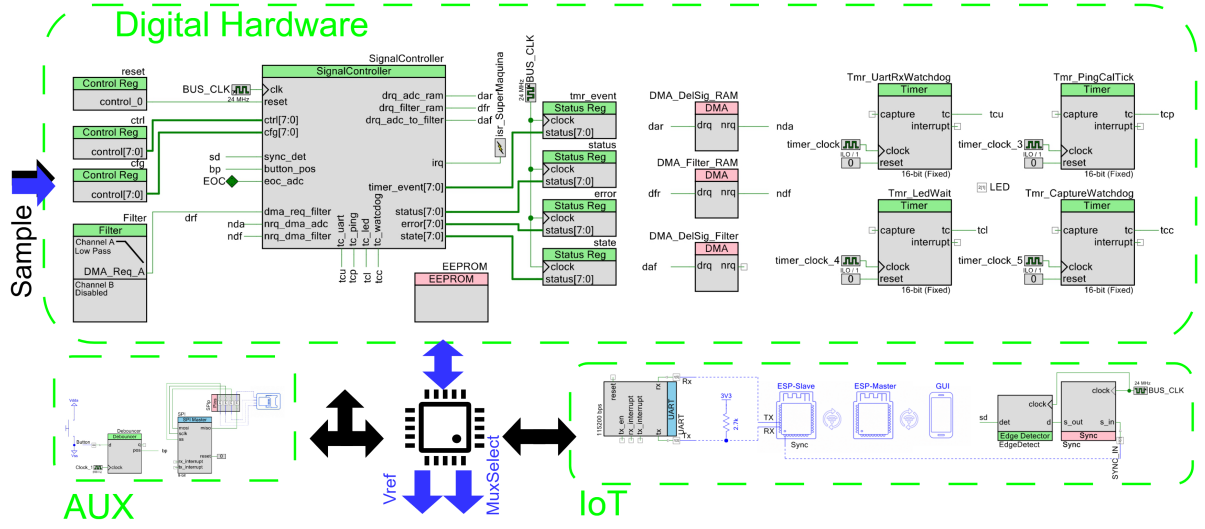


Fig. 3. Digital calibration hardware. Direct memory access (DMA) moves ADC/filter data through RAM; the controller reconfigures the analog multiplexer, while firmware accesses EEPROM. Black and blue arrows indicate acquisition and calibration paths, respectively.

the valid-stage mask, four DAC codes, version, and cyclic-redundancy check (CRC). A failed run may hold its bounded best codes for the current sequence but never overwrites a valid record.

B. Control Law and Calibration Parameters

For each stage, the ADC reports the node voltage relative to V_{ref} . Thus, $\hat{y}_i[n] = 0$ denotes a physical node at V_{ref} , not at 0 V. The variables $\hat{y}_i[n]$, r_i , and $e_i[n]$ are signed ADC counts; the filtered differential measurement gives $e_i[n] = r_i - \hat{y}_i[n]$, with $r_i = 0$. The error is first mapped to the 8-bit DAC domain:

$$\begin{aligned} \varepsilon_i[n] &= \text{round}\left(e_i[n] \frac{q_{\text{ADC}}}{q_{\text{DAC}}}\right) \\ &= \text{round}\left(\frac{e_i[n] V_{\text{ADC,span}} N_{\text{DAC}}}{N_{\text{ADC}} V_{\text{DAC,span}}}\right), \end{aligned} \quad (2)$$

where $q_x = V_{x,\text{span}}/N_x$ is the step size and N_x is the number of quantization levels, not the maximum code. Here $N_{\text{ADC}} = 2^{18}$ and $N_{\text{DAC}} = 2^8$. At nominal $V_{\text{DDA}} = 5$ V, $V_{\text{ADC,span}} = 5000$ mV and $q_{\text{ADC}} = 5000/2^{18}$ mV/count; physically, the differential span follows V_{DDA} . The VDAC's ideal 256-bin span is 4096 mV, so $q_{\text{DAC}} = 16$ mV/code, while code 255 reaches 4080 mV [7], [15].

The rounding in Eq. 2 gives an integer error in equivalent DAC codes. An overly narrow band can alternate indefinitely between adjacent codes when the ideal point is not realizable; the implementation therefore adopts the conservative one-step floor

$$\Delta_{i,\text{min}}^{\text{impl}} = \max(1, \lceil |K_i| \rceil), \quad (3) \quad \bar{\varepsilon}_i[n] = \begin{cases} 0, & |\varepsilon_i[n]| \leq \Delta_i, \\ \varepsilon_i[n], & \text{otherwise,} \end{cases} \quad (4)$$

Inside the band, Eq. 4 suppresses integration; lock and refinement still govern convergence. Table I lists adopted values satisfying $\Delta_i \geq \Delta_{i,\text{min}}^{\text{impl}}$.

Under the assumptions of Section II, the architecture requires a measurable stage output and sufficient trim authority and

TABLE I
PARAMETERS USED FOR THE REPORTED MEASUREMENTS. DEADBANDS ARE IN EQUIVALENT DAC CODES; S/L/R ARE SETTLE/LOCK/REFINE SAMPLES AT 2604 HZ, WHEREAS TIMEOUT IS IN SECONDS.

Stage	$ K_i $	$\Delta_{i,\text{min}}^{\text{impl}}$	Δ_i	$k_{P,i}$	$k_{I,i}$	S/L/R	Timeout (s)
PGA	1	1	18	10^{-3}	3×10^{-4}	512/512/1024	11.5
BP	1	1	3	10^{-4}	5×10^{-4}	512/1024/1024	17.3
SUM	7.89	8	10	10^{-4}	5×10^{-4}	512/512/1024	11.5
LP	6	6	8	10^{-4}	5×10^{-4}	1024/1024/2048	17.3

range. The controller is written in positional proportional-integral (PI) form,

$$c_i^*[n] = c_{i,0} + s_i \left[\frac{k_{P,i}}{|K_i|} \bar{\varepsilon}_i[n] + \frac{k_{I,i}}{|K_i|} \sum_{q=0}^n \bar{\varepsilon}_i[q] \right], \quad (5)$$

Here $c_{i,0}$ is a feedforward code near the expected operating point and, under the adopted error convention, $s_i = \text{sgn}(K_i)$. The four physical K_i values in Table I are positive, so $s_i = +1$. The gain magnitudes follow from the topology and resistor ratios; the gain-2 two-op-amp input of [14], followed by the programmable-gain block (Fig. 1), gives nominal differential signal gain $2 \text{Gain}_{\text{PGA}}$, whereas its DC VDAC-to-node gain is $K_{\text{PGA}} = 1$. The reported runs used $c_0 = (156, 156, 156, 156)$ for (PGA, BP, SUM, LP) and the $k_{P,i}$ and $k_{I,i}$ values in Table I.

The command is clamped and slew-limited to one code per sample, with conditional integration for anti-windup. Convergence requires a fixed sample count in one quantized bucket; an adjacent-code trial is retained only if it lowers the filtered error. At timeout, the best code is held, the stage is flagged, and the sequence advances. Codes are stored only if all controllable stages succeed; otherwise the prior EEPROM record remains.

The proof-of-concept parameter set in Table I was fixed across all reported conditions and was not systematically optimized; Section V-A discusses $\Delta_{\text{PGA}} = 18$. Finer trim

can use a current-output DAC/resistor or buffered topology [15] without changing the sequence.

IV. PROOF-OF-CONCEPT EVALUATION

A. Prototype and Comparison Conditions

Two independently assembled SM-24 prototypes (Boards A and B) were compared under a *shared fixed-reference baseline*, $c_{\text{base}} = (156, 156, 156, 156)$ for (PGA, BP, SUM, LP), selected on Board B at $\times 50$ and reused at every gain, and *Calibrated*, four-stage calibration followed by frozen-code acquisition. The measurement firmware disabled post-calibration EEPROM writes, so every stationary-geophone restart reloaded that vector. Thus, $N=10$ denotes paired restarts of one board/gain configuration, not independent devices. Normal verification made no corrective update on Board B $\times 1$ or $\times 4$, producing the dashes in Table II. A separate $\times 4$ diagnostic forced three paired updates with all $\Delta_i = 1$ and is excluded because its bands and sample count differ.

B. Residual DC-Offset Measurement and Results

Offsets are nominal 5-V full-scale fractions, not values normalized to instantaneous supply voltage. LP is the primary system outcome because it faces the acquisition ADC; PGA, BP, and SUM document sequential correction and expose near-rail intermediate operating points.

At each setting, a four-channel Tektronix TDS 2004B 8-bit, 60 MHz oscilloscope recorded V_{mean} at PGA, BP, SUM, and LP over a 100 ms post-settling window. All four probe commons were tied to $V_{\text{ref}} = V_{\text{DDA}}/2$, so channels measured $V_{\text{stage}} - V_{\text{ref}}$. PC-USB-supplied V_{DDA} was typically below 5 V. The unearthed PC and protective-earth-referenced oscilloscope shared a mains branch, so the scope intentionally earth-referenced V_{ref} [18]; other leakage paths were not characterized. Recording baseline before calibration preserved within-restart pairing, but the fixed order leaves slow drift as a possible confound.

Table II reports the mean and sample standard deviation of $100|V_{\text{mean}}|/5\text{ V}$ over ten paired restarts. The largest fixed-reference deviations approached a supply rail; because per-run supply and output-headroom margins were not logged, these are observed output deviations and may be lower bounds on the underlying linear offset. These descriptive statistics characterize within-configuration repeatability; no population-level inferential test or independent oscilloscope uncertainty budget was performed, and displayed digits do not imply sub-millivolt accuracy.

Controller acceptance uses a filtered internal ADC estimate; Table II uses a later, asynchronous scope record without logged final ADC errors or flags. The scope residuals therefore do not directly verify Δ_i , and bandwidth, elapsed-time, and residual BP-capacitor-state effects after upstream updates were not separated. Fig. 4's colored traces were produced offline by passing the scope records through an equivalent FIR; they are not logged controller states.

Across the eight reported board $\times$ gain cells, the calibrated LP sample mean is 0.4–3.9%FS versus 2.9–45.8%FS under

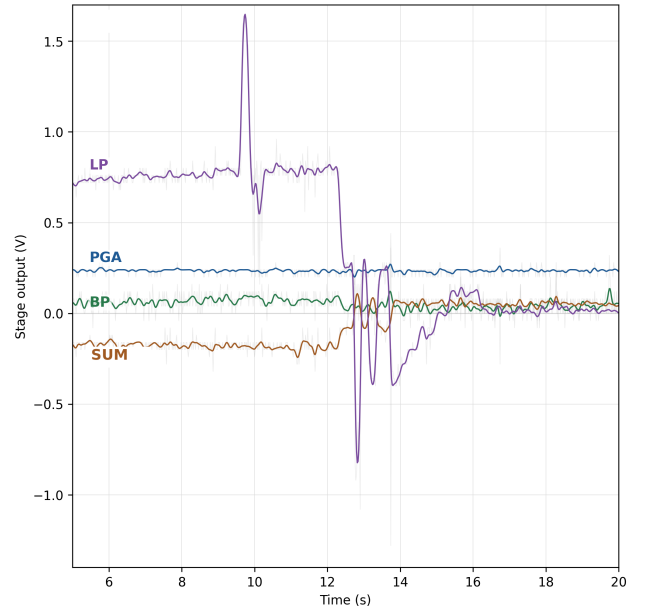


Fig. 4. Stage deviations relative to V_{ref} during calibration at setting $\times 16$ (total differential gain $\times 32$). Thin gray traces are oscilloscope records; colored traces are their offline outputs from an equivalent FIR. See Section V-A.

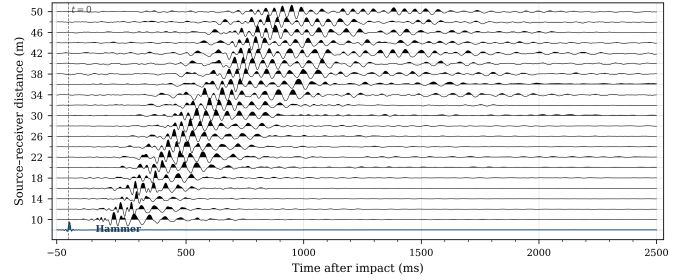


Fig. 5. Wiggle-trace field record acquired after foreground calibration. The trigger and first-arrival moveout demonstrate multichannel operation over the 10–50 m line; this is not a controlled fixed-reference comparison.

the shared fixed-reference baseline; individual calibrated observations span 0.1–5.2%FS (Table II). The LP mean was lower after calibration in every reported cell, without board- or gain-specific controller retuning.

C. Field Operation Demonstration

This field trial was an operational demonstration, not a controlled offset or surface-wave comparison. Fig. 5 shows a 10–50 m record; each geophone was planted and kept stationary during the preceding calibration. During informal field operation, a timeout was observed when a board was powered while its geophone was moving: the AC transient saturated the chain and violated the quasi-static assumption. The resolved first-arrival moveout demonstrates operation.

V. DISCUSSION

Unlike precision front ends characterized near the sensor's Johnson-noise limit [16], this method trades component precision for measurement and digital assistance. Unlike software subtraction, it physically recenters observable additive DC errors within DAC authority and range. The comparison is

TABLE II
RESIDUAL DC OFFSET (%FS OF THE NOMINAL 5-V ADC SPAN), MEAN $\pm$ SAMPLE STANDARD DEVIATION OVER $N=10$ PAIRED BASELINE-SEEDED RESTARTS PER REPORTED BOARD/GAIN CELL.

PGA setting	Condition	Board A				Board B			
		PGA	BP	SUM	LP	PGA	BP	SUM	LP
$\times 1$	Fixed reference	3.5 ± 0.2	1.5 ± 2.2	$21.8 \pm 1.1^\dagger$	$45.8 \pm 3.1^\dagger$	—	—	—	—
	Calibrated	3.4 ± 0.3	1.2 ± 0.2	0.8 ± 0.2	3.2 ± 0.9	—	—	—	—
$\times 4$	Fixed reference	2.4 ± 0.1	4.9 ± 0.3	6.7 ± 1.0	$42.6 \pm 4.2^\dagger$	—	—	—	—
	Calibrated	2.1 ± 0.3	1.0 ± 0.2	1.0 ± 0.3	3.9 ± 0.8	—	—	—	—
$\times 16$	Fixed reference	2.4 ± 0.1	3.8 ± 0.9	4.9 ± 1.5	$29.6 \pm 6.8^\dagger$	2.1 ± 0.1	0.66 ± 0.02	4.5 ± 0.8	3.0 ± 0.1
	Calibrated	1.9 ± 0.2	1.5 ± 1.5	1.4 ± 1.5	2.3 ± 1.4	1.7 ± 0.1	0.66 ± 0.02	0.6 ± 0.5	0.7 ± 0.1
$\times 32$	Fixed reference	1.0 ± 0.2	7.0 ± 0.4	7.4 ± 0.3	$12.4 \pm 1.3^\dagger$	1.0 ± 0.2	0.66 ± 0.01	8.0 ± 0.8	2.9 ± 0.1
	Calibrated	0.5 ± 0.2	1.0 ± 0.5	0.6 ± 0.2	1.7 ± 0.7	0.5 ± 0.2	0.66 ± 0.01	1.2 ± 0.3	0.4 ± 0.2
$\times 50$	Fixed reference	1.2 ± 0.3	1.0 ± 0.7	0.4 ± 0.2	4.1 ± 2.1	2.5 ± 0.1	0.65 ± 0.01	0.9 ± 0.6	2.9 ± 0.1
	Calibrated	1.0 ± 0.5	0.9 ± 0.5	$0.8 \pm 0.6^\ddagger$	1.2 ± 0.4	1.8 ± 0.3	0.65 ± 0.02	0.9 ± 0.6	0.7 ± 0.2

Lower is better. † Fixed reference $\geq 10\%$ FS. ‡ The marked calibrated mean is larger than its baseline; equality after rounding is not marked (Section V-A). Dashes denote cells in which normal verification made no corrective update, so no $N=10$ comparison was collected. N denotes restarts of the same configuration, not independent boards; the separate Board B $\times 4$ three-pair diagnostic is excluded.

against one shared fixed-reference baseline, not optimized per-condition fixed codes, and evaluates the complete sequence rather than each trim; per-stage ablation remains future work.

Continuous DC-servo loops introduce a settling-versus-bandwidth tradeoff [17]; the foreground controller instead holds constant DAC outputs during acquisition and avoids a continuously active feedback path. Limitations include finite DAC resolution, no noise or distortion correction, and unvalidated convergence under sustained vibration, thermal drift, or slow internal filter-state transients.

A. Non-Optimized PGA Acceptance

PGA offset is only modestly smaller than baseline. With $|K_{PGA}| = 1$, the non-optimized $\Delta_{PGA} = 18$ accepts integer-error buckets centered through ± 288 mV; because error is rounded first, the continuous nominal edge is approximately ± 296 mV. The 0.24-V level in Fig. 4 lies inside it, consistent with stop-on-band rather than instability, but does not establish an accuracy floor. In the separate Board B $\times 4$ diagnostic, all $\Delta_i = 1$ forced updates and reduced three external LP residuals from 148/142/157 mV to 34.4/38.5/53.5 mV. Because all bands changed and only three pairs were collected, this result is exploratory.

The only increase visible at the reported precision, Board A SUM at $\times 50$ (0.4%FS to 0.8%FS), is compatible with band acceptance and does not affect the downstream LP result.

B. Timing and Scope

Visible correction in Fig. 4 spans about 7 s. An unlogged bench observation placed completion near 10 s, but runs were not systematically timed. The four timeout windows sum to 57.6 s; adding the listed settling and refinement gives approximately 60.5 s, excluding stored-code verification and software/switching overhead. No laboratory timeout was observed.

One initialization selected on Board B at $\times 50$ transferred to Board A and the reported gains. However, two boards cannot establish population scaling and fixed ordering leaves drift as a confound. Future work requires more independent boards with randomized order, temperature and parameter-sensitivity studies, per-stage ablation, and controlled field comparisons.

VI. CONCLUSION

In this two-board proof of concept, the ADC-facing LP sample-mean offset was 0.4–3.9%FS after four-VDAC calibration versus 2.9–45.8%FS under the shared fixed-reference baseline across eight reported cells, with codes frozen during acquisition. The 10–50 m field record confirms integration, not comparative performance. Future work will expand the independent hardware population, sensitivity and ablation studies, and controlled field evaluation.

ACKNOWLEDGMENT

The authors used Anthropic Claude and OpenAI Codex for language drafting, structural revision, and consistency checks in Sections I–VI. The authors independently verified and take responsibility for all technical claims, calculations, measurements, figures, and references.

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